- Rec Sys
  - Collaborative Filtering
    - User - User
    - Item - Item
  - Content - based
  - Rec as Classification/ Regression

- → Amazon (Sales | Revenue)
- → Netflix (Retention, Watchtime)
- → Instagram / Tiktok / Linkedin (Watchtime)

Pre 2007

- → Content Based
- → Collaborative filtering
  - → Item - Item
  - → User - User

2008 - 2015

→ Matrix Factorization (Netflix)

2016 – Current

→ Deep Learning.

User : u_i $\quad i : 1 \rightarrow n$ $\quad \longrightarrow u_1, u_2 \text{ --- } u_n$

Billions

Items : I_j $\quad j : 1 \rightarrow m$ $\quad \longrightarrow I_1, I_2 \text{ --- } I_m$

Millions

Objective : $u_i \rightarrow I_{10}, I_{20}, I_5, I_{18}$ (Recommend)

1. Explicit Rating → User rates it

→ Amazon [1 - 5]

→ Netflix [0, 1]

2. Implicit Rating → 50% → 1

→ 0

50%

1hr

1/0

user	Item	Rating
u_1	I_1	5
u_1	I_{10}	3
u_2	I_2	2
u_2	I_9	5
⋮	⋮	⋮

User-Item matrix

	I_1	I_2	$\cdots$	I_9	I_{10}	$\cdots$	I_m
u_1	5	0	—		3	—	—
u_2	—	2		5		—	—
⋮	—	2	—	1		—	—
u_n							

$u_2 \quad I_2 \text{ — } 2$

$n = 10^9$

$m = 10^6$

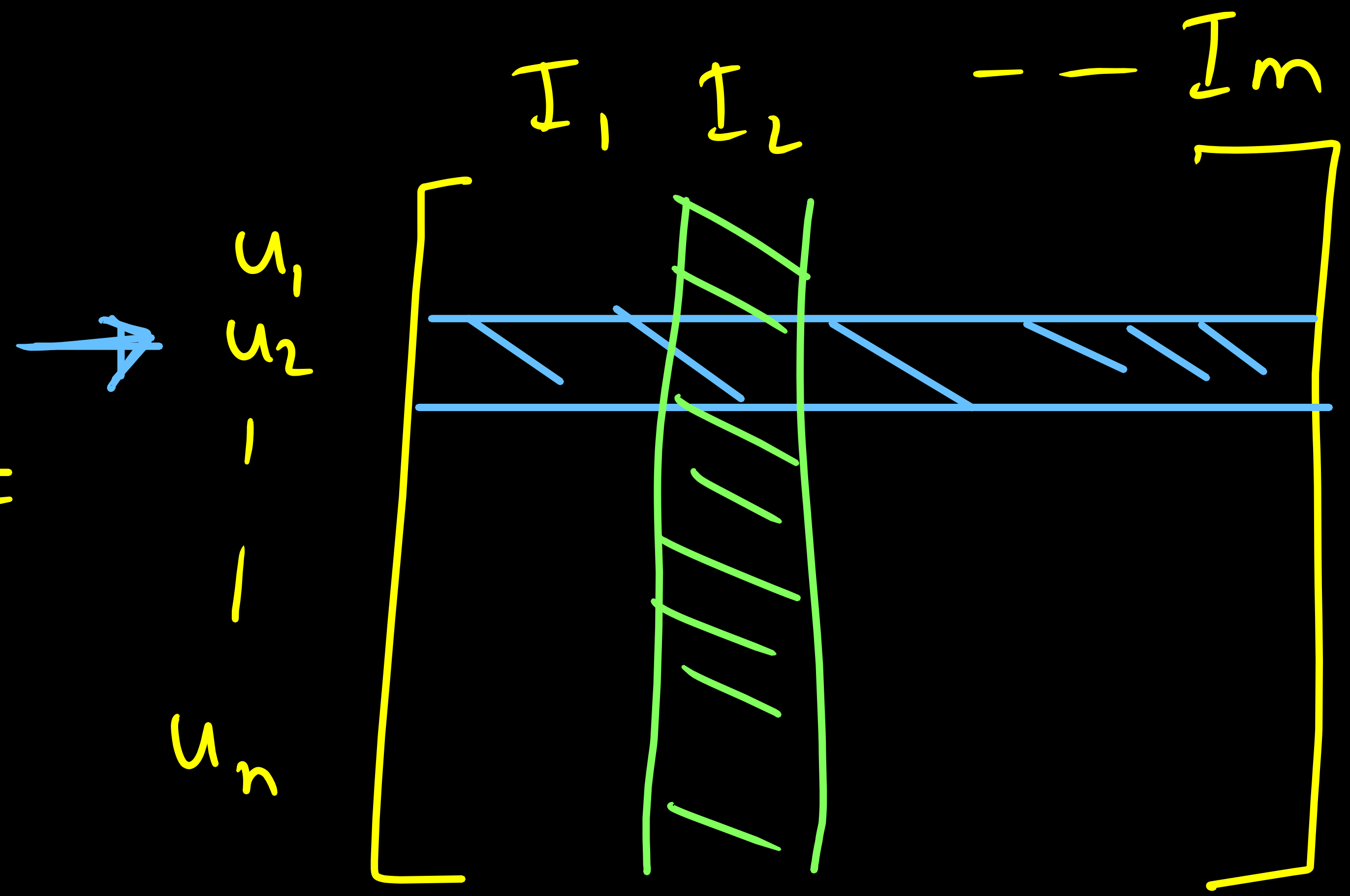
$n \times m = 10^{15}$

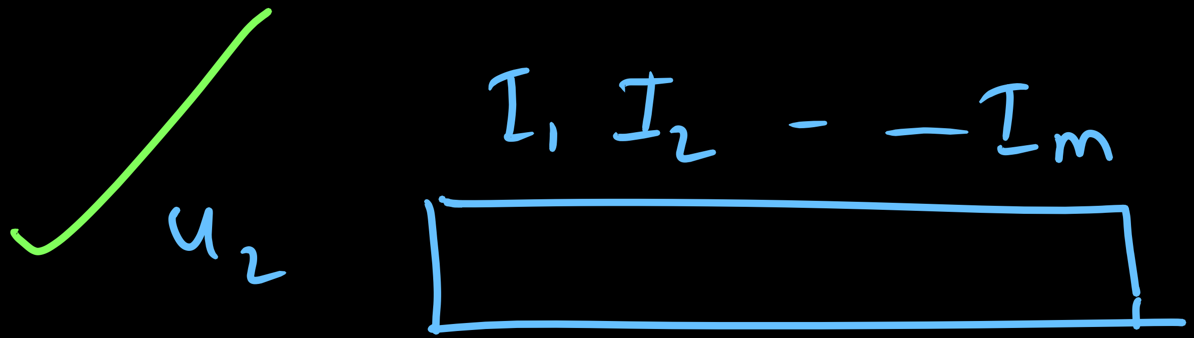
$$\text{Sparsity} = \frac{10^9 \times 100}{n \times m} = \frac{10^{11}}{10^{15}} = 10^{-4}$$

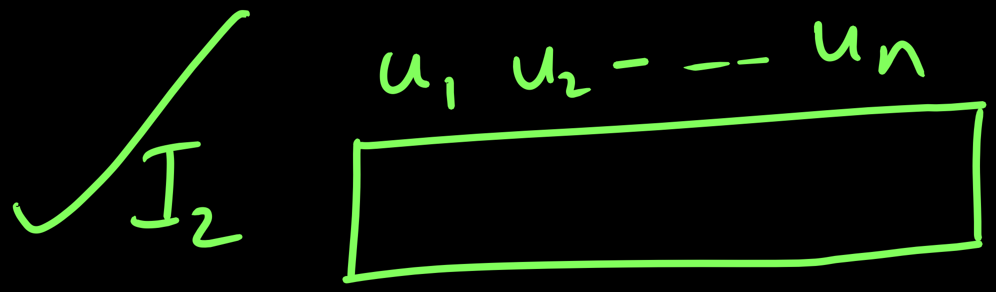
→ Collaborative filtering → Item-Item

<u>User-Item</u>

$A_{n \times m} =$







→ U_{10} has watched movie I_1, I_3, I_7

↳ Recommend

$I_1 \rightarrow \{ I_4, I_{10} \}$

$I_3 \rightarrow \{ I_4, I_6 \}$

$I_7 \rightarrow \{ I_{10}, I_1 \}$

→ top-2 similar movie

I_4, I_{10}

User - Item

$$A_{n \times m} = \begin{array}{c|cccc} & I_1 & I_2 & \cdots & I_m \\ \hline u_1 & & & & \\ u_2 & & & & \\ \vdots & & & & \\ u_n & & & & \end{array}$$

$$\cos(I_i, I_j) = \frac{I_i \cdot I_j}{\|I_i\| \, \|I_j\|}$$

$$= \hat{I}_i \cdot \hat{I}_j$$

1) Normalize according to columns

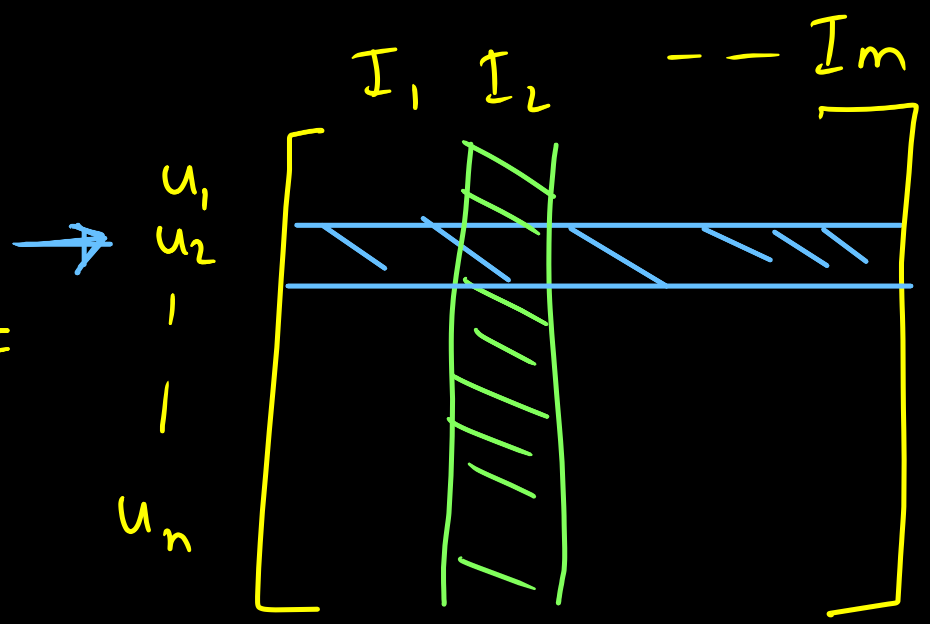
2) $\underset{m \times n}{A^T} \, \underset{n \times m}{A} = \underset{n \times n}{\text{Item-Item}}$

$$
\begin{array}{c|c}
 & \begin{array}{cccc} I_1 & I_2 & I_3 & I_4 \end{array} \\
\begin{array}{c} I_1 \\ I_2 \\ I_3 \\ I_4 \end{array} & \begin{bmatrix} 1 & 0.9 & 0.4 & 0.2 \\ & & & \\ & & & \\ & & & \end{bmatrix}_{m \times m}
\end{array}
$$

User-User Similarity

User-Item

$A_{n \times m} =$



$$\text{cosine}(u_i, u_j) = \frac{u_i \cdot u_j}{\|u_i\| \|u_j\|}$$

$$= \hat{u}_i \cdot \hat{u}_j \longrightarrow$$

1> Normalize according to row

2> $\underset{n \times m}{A}\underset{m \times n}{A^T} = \underset{n \times n}{\text{User-User}}$

u_i has brought item I_{10} , I_{18}

top-3

$u_{10} \rightarrow \{ \cancel{I_{10}} , \overset{5}{I_{12}} , \cancel{I_{18}} , \overset{4}{I_{20}} \}$

$u_{26} \rightarrow \{ \cancel{I_{10}} , \cancel{I_{18}} , \overset{3}{I_{26}} , I_{12} \} \rightarrow 4$

$u_{58} \rightarrow \{ \cancel{I_{18}} , I_{12} \} \rightarrow 5$

$u_i \rightarrow \{ \overset{\frac{5+4+4}{3}}{I_{12}} , \overset{4}{I_{20}} , \overset{3}{I_{26}} \}$

→ Item - Item or User - User

1. $n \gg m$ Item - Item $m \times m$ User - User $n \times n$

2. Item preference does not change

Cold Start

→ New user

↳ Vector is empty as no rating

User - Item

$$A_{n \times m} = \begin{array}{c} \\ u_1 \\ u_2 \\ | \\ | \\ u_n \end{array} \begin{array}{c} \begin{array}{cccc} I_1 & I_2 & -- & I_m \end{array} \\ \left[\begin{array}{cccc} & & & \\ & & & \\ & & & \\ & & & \\ & & & \end{array} \right] \end{array}$$

→ New Item

↳ Vector is empty as no rating

Content-based

→ New user

- Location ✓
- Gender ✓
- Age ✓
- Genre ✓
- Language ✓

New Item/Movie

↳ ~~Star Cast~~

↳ Director

↳ Genre

↳ Language

→ top-5

top-10

→ Break until 22:25 PM

1) Movie represented by features like director, genre etc

Media Attributes

	Director	Genre	Tags
Inception	Christopher Nolan	Action, Adventure, Sci fi	Psychological manipulation, Mind bender ...
Interstellar	Christopher Nolan	Action, Adventure, Drama	Time paradox ...
Baby's day out	Patrick Read Johnson	Drama, Comedy ...	1990s

Attributes are preprocessed (stop word removed, words converted to lowercase, tags with multiple words joined) to extract features

Attribute Preprocessing

Extracted features

	Patrick Read Johnson	Christopher Nolan	Action	Adventure	Sci fi	Drama	Comedy	Psychological manipulation	Mind Bender	Time Paradox	1990s
Inception	0	1	1	1	0	0	1	1	0	1	
Interstellar	1	0	1	1	1	0	0	0	1	0	
Baby's day out	0	0	1	0	1	1	0	0	0	1	

Term frequencies (Binary Representation)

BoW / TF-IDF

2> Build a user profile

↳ User Metadata [Cold start]

↳

TF-IDF/BoW

Creating user A's user profile
Considering only 3 movies instead of the whole corpus for simplicity

Movie profiles

	Patrick Read Johnson	Christopher Nolan	Action	Adventure	Sci fi	Drama	Comedy	Psychological manipulation	Mind Bender	Time Paradox	1990s
Inception	0.00	0.21	0.12	0.06	0.09	0.00	0.00	0.16	0.21	0.00	0.00
Interstellar	0.00	0.23	0.00	0.07	0.10	0.14	0.00	0.00	0.00	0.23	0.00
Baby's day out	0.31	0.00	0.00	0.07	0.00	0.14	0.14	0.00	0.00	0.00	0.23

user historical interaction watched (1) or ignored (-1)

User A
1
0
-1

Interstellar has never been recommended to user A. it is a new movie in the system

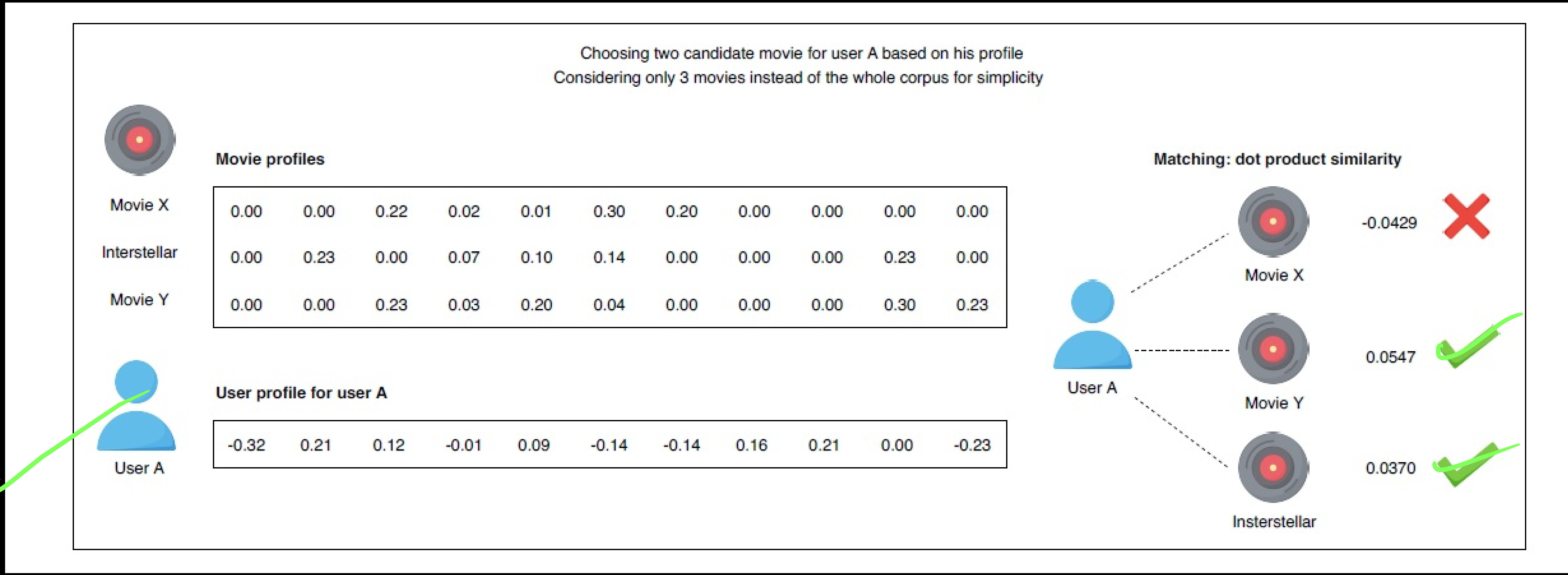
User profile for user A

-0.31	0.21	0.12	-0.01	0.09	-0.14	-0.14	0.16	0.21	0.00	-0.23

User A

$(0 \times 1) + (0 \times 0) + (0.23 \times -1)$

3) Find the similarity of user profile w.r.t movies



Movie X	0.00	0.00	0.22	0.02	0.01	0.30	0.20	0.00	0.00	0.00	0.00
Interstellar	0.00	0.23	0.00	0.07	0.10	0.14	0.00	0.00	0.00	0.23	0.00
Movie Y	0.00	0.00	0.23	0.03	0.20	0.04	0.00	0.00	0.00	0.30	0.23

User A	-0.32	0.21	0.12	-0.01	0.09	-0.14	-0.14	0.16	0.21	0.00	-0.23

$$\text{Cosine}(u_i \cdot I_j) = \frac{u_i \cdot I_j}{\|u_i\| \, \|I_j\|}$$

→ Content based

1> Content based on feature like Age, gender, Genre (Domain knowledge)

2> Can handle cold start

3> Does not require info about other users

4> Cannot discover new interest

→ Collaborative → Item - Item
→ User - User

1> No domain knowledge

2> Cannot handle cold start

3> Requires

4> Can discover new interest

u_1 → Romantic, Scifi

u_2 → Scifi, Comedy

→ Hybrid (Content + Collaborative)